# Observer-Dependent Dark Matter: Gravitational Entanglement and the MOND Scale from Quantum Retrodiction

Sheng-Kai Huang[1, *]

[1] *Independent Researcher*

(Dated: March 12, 2026)

We propose that dark matter phenomenology at galactic scales arises from the observer-dependence of gravitational entropy production $\Sigma$. A classical observer, with access only to electromagnetic degrees of freedom, assigns a larger entropy production $\Sigma_{\rm cl}$ to a gravitational process than a quantum observer who can additionally access the gravitational entanglement structure: $\Sigma_{\rm cl} \geq \Sigma_{\rm q}$, by the data processing inequality. The difference $\Sigma_{\rm DM} \equiv \Sigma_{\rm cl} - \Sigma_{\rm q}$ manifests as apparent dark matter. We develop this framework quantitatively by connecting observer-dependent $\tau = 1 - F$ (Paper V) to the scale-dependent gravitational channel: the anomalous dimension $\eta = 1$ identified by Kumar [arXiv:2509.05246], reproduced within the $\tau$ framework via the data processing inequality and extensivity, produces a logarithmic correction to the Newtonian potential, yielding flat rotation curves with a universal crossover scale $k_* \approx 2.7 \times 10^{-2}$ kpc$^{-1}$, consistent with the 100-galaxy SPARC validation of Gubitosi *et al.* The MOND acceleration scale $a_0 \approx cH_0/(2\pi)$ is related to the KMS thermal periodicity of the de Sitter vacuum through the Crooks fluctuation theorem, interpreting the MOND transition as the onset of marginal irreversibility ($\Sigma \sim 1$). A Crooks-inspired interpolation function $\mu(x) = 1 - e^{-x}$ with $x = a/a_0$ is proposed and shown to be competitive with standard MOND interpolations. The Bullet Cluster is addressed through an entanglement rigidity argument. The Khronon coupling $\mu_0 = H_0/c$—the unique dimensionally-determined value from the de Sitter temperature—with anomalous dimension $\eta_\mu = 1$ produces dust-like behavior ($\tilde{w}_0 \sim 6 \times 10^{-11}$) at CMB scales, providing a candidate for the pressureless mode needed for the acoustic peaks. The exact $\Omega_{\rm DM} h^2$ prefactor requires the numerical computation of Paper IV, and this CMB resolution remains unverified.

## I. INTRODUCTION

The dark matter problem is one of the central puzzles in physics. Galaxy rotation curves flatten at large radii [1], the cosmic microwave background (CMB) acoustic peaks require a pressureless, non-baryonic component with $\Omega_{\rm CDM} h^2 \approx 0.12$ [2], and gravitational lensing in the Bullet Cluster shows mass displaced from the dominant baryonic component [3]. These observations are well described by the $\Lambda$CDM concordance model, in which cold dark matter (CDM) particles constitute roughly 27% of the energy budget of the universe.

Yet four decades of direct detection experiments have found no CDM particles [4]. This null result, combined with the tight empirical scaling relations that CDM does not predict—the radial acceleration relation (RAR) [5], the baryonic Tully–Fisher relation [6], and the near-universal acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s$^2$—motivates the search for alternative explanations.

Modified Newtonian dynamics (MOND) [7] reproduces galactic phenomenology with a single parameter $a_0$ but lacks a satisfactory relativistic completion and fails at CMB scales [8]. Verlinde's emergent gravity [9] derives $a_0 \sim cH_0/6$ from volume-law entanglement in de Sitter space and achieves zero-parameter fits to rotation curves, but has no covariant formulation and fails quantitatively at cluster scales [10]. The AeST theory of Skordis and Zlošnik [8] fits both galactic rotation curves and the CMB power spectrum, but does so by introducing a vector field and scalar field that are, in a precise sense, functionally equivalent to a dark sector.

In this paper, we propose a different perspective. We argue that "dark matter" at galactic scales is the gravitational information that classical observers cannot access. The framework rests on the observer-dependent temporal asymmetry parameter $\tau_O = 1 - F(\rho, \mathcal{R} \circ \mathcal{N}_O(\rho))$ developed in Ref. [11] (hereafter Paper V), where $F$ is the fidelity of Petz recovery and $\mathcal{N}_O$ is the effective channel determined by the observer's accessible algebra $\mathcal{A}_O$. A classical astronomer, restricted to electromagnetic observables ($\mathcal{A}_{\rm cl}$), assigns a larger entropy production $\Sigma_{\rm cl}$ to the same gravitational process than a hypothetical quantum observer with access to the full gravitational entanglement structure ($\mathcal{A}_{\rm q} \supseteq \mathcal{A}_{\rm cl}$). The data processing inequality (DPI) guarantees $\Sigma_{\rm cl} \geq \Sigma_{\rm q}$, and the excess $\Sigma_{\rm DM} = \Sigma_{\rm cl} - \Sigma_{\rm q}$ manifests as apparent dark matter.

The paper is organized as follows. Section II formalizes observer-dependent dark matter using the subalgebra framework of Paper V. Section III develops the galactic-scale phenomenology: running Newton's constant from the renormalization group (RG) flow of the gravitational channel, flat rotation curves, the RAR, and the baryonic Tully–Fisher relation. Section IV relates the MOND acceleration scale to the Crooks fluctuation theorem in de Sitter space and proposes a Crooks-inspired interpolation function. Section V addresses the Bullet Cluster. Section VI discusses cosmic acceleration and dark energy. Section VII assesses the CMB gap and identifies the structural parallel with AeST/Khronon theories; the

* akai@fawstudio.com

running Khronon coupling $\mu_0 = H_0/c$ with $\eta_\mu = 1$ then provides a resolution yielding dust-like behavior at CMB scales, with the full treatment deferred to Paper IV. Section VIII lists testable predictions, and Sec. IX provides a critical assessment of what is genuinely new and what remains open.

## II. FRAMEWORK: OBSERVER-DEPENDENT DARK MATTER

### A. Review: observer-dependent $\tau$

In Paper V [11], each observer $O$ is identified with a von Neumann subalgebra $\mathcal{A}_O \subseteq \mathcal{A}_{\text{total}}$ of the total algebra of observables. The observer's effective channel is the conditional expectation $\mathcal{N}_O = \mathrm{Tr}_{\mathcal{A}_O^c}$, and the observer-dependent temporal asymmetry is

$$\tau_O = 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}_O}(\mathcal{N}_O(\rho))\big), \tag{1}$$

bounded by the JRSWW inequality [12]:

$$\tau_O \leq 1 - \exp(-\Sigma_O/2), \tag{2}$$

where $\Sigma_O = D(\rho\|\sigma) - D(\mathcal{N}_O(\rho)\|\mathcal{N}_O(\sigma))$ is the relative entropy lost by restricting to $\mathcal{A}_O$. The monotonicity theorem (Theorem 1 of Paper V) states: if $O_1 \leq O_2$ (i.e., $\mathcal{A}_{O_1} \subseteq \mathcal{A}_{O_2}$), then $\tau_{O_1} \geq \tau_{O_2}$ and $\Sigma_{O_1} \geq \Sigma_{O_2}$.

### B. Classical and quantum observers for a galaxy

**Definition 1** (Classical observer)**.** *The classical galactic observer $O_{\text{cl}}$ has accessible algebra*

$$\mathcal{A}_{\text{cl}} = \mathcal{A}_{\text{EM}} \otimes \mathbf{1}_{\text{grav}} \otimes \mathbf{1}_{\text{env}}, \tag{3}$$

*where $\mathcal{A}_{\text{EM}}$ contains all observables constructible from electromagnetic field modes and their couplings to baryonic matter (photometry, spectroscopy, 21 cm emission, thermal bremsstrahlung).*

**Definition 2** (Quantum observer)**.** *The quantum galactic observer $O_{\text{q}}$ has*

$$\mathcal{A}_{\text{q}} = \mathcal{A}_{\text{EM}} \vee \mathcal{A}_{\text{grav}}, \tag{4}$$

*the algebra generated by $\mathcal{A}_{\text{EM}}$ and the gravitational entanglement observables $\mathcal{A}_{\text{grav}}$, which include vacuum entanglement between spatial regions of the gravitational field, volume-law entanglement from the de Sitter background [9], and modular flow observables of the gravitational field [16].*

Since $\mathcal{A}_{\text{cl}} \subset \mathcal{A}_{\text{q}}$, the monotonicity theorem gives:

**Theorem 1** (Dark matter inequality)**.** *For any gravitational process with reference state $\sigma$,*

$$\Sigma_{\text{cl}} \geq \Sigma_{\text{q}} \geq 0, \tag{5}$$

*with equality if and only if the gravitational entanglement carries no information beyond what is accessible electromagnetically.*

*Proof.* Immediate from the DPI applied to the conditional expectation $\mathcal{E} : \mathcal{A}_{\text{q}} \to \mathcal{A}_{\text{cl}}$, since $\mathcal{N}_{\text{cl}} = \mathcal{E} \circ \mathcal{N}_{\text{q}}$. □

**Definition 3** (Dark matter entropy production)**.** *The observer-dependent dark matter entropy production is*

$$\Sigma_{\text{DM}} \equiv \Sigma_{\text{cl}} - \Sigma_{\text{q}} \geq 0. \tag{6}$$

### C. From $\Sigma_{\text{DM}}$ to apparent dark matter mass

Using the connection $\Sigma_{\text{grav}} = -\ln(-g_{00})$ established in Ref. [13] (Paper II), the gravitational potential is $\Phi(r) = -(c^2/2)\,\Sigma(r)$ in the weak-field limit. The mass inferred by the classical observer within radius $r$ is

$$M_{\text{obs}}(r) = \frac{c^2 r^2}{2G_N}\,\Sigma'_{\text{cl}}(r), \tag{7}$$

and the baryonic mass attributed by the quantum observer is

$$M_{\text{bar}}(r) = \frac{c^2 r^2}{2G_N}\,\Sigma'_{\text{q}}(r). \tag{8}$$

The apparent dark matter mass is therefore

$$M_{\text{DM}}(r) = M_{\text{obs}}(r) - M_{\text{bar}}(r) = \frac{c^2 r^2}{2G_N}\,\Sigma'_{\text{DM}}(r). \tag{9}$$

The challenge is to determine $\Sigma_{\text{DM}}(r)$. We now show that the scale-dependent gravitational channel provides the answer.

## III. GALACTIC SCALE: ROTATION CURVES

### A. Scale-dependent gravitational channel

The gravitational field acts as a quantum channel $\mathcal{N}_{\text{grav}}$ that processes quantum information. At different momentum scales $k$ (corresponding to distance scales $r \sim 1/k$), the channel has different properties because Newton's constant $G$ runs under the renormalization group [18, 19]. The entropy production at scale $k$ is

$$\Sigma(k) = D(\rho\|\sigma) - D\big(\mathcal{N}_{\text{grav}}(k)(\rho)\big\|\mathcal{N}_{\text{grav}}(k)(\sigma)\big). \tag{10}$$

The DPI, applied to the RG flow—which is itself a quantum channel [14]—gives a monotonicity condition:

$$\Sigma(k_{\text{IR}}) \geq \Sigma(k_{\text{UV}}) \geq 0. \tag{11}$$

The gravitational channel at infrared (IR) scales is noisier—it loses more information—than at ultraviolet (UV) scales. This monotonicity is the entropic $c$-theorem of Casini and Huerta [15], reinterpreted as a statement about the gravitational channel.

### B. Running $G$ and the logarithmic correction

The anomalous dimension $\eta$ governs the IR running:

$$\frac{d\ln G}{d\ln k} = -\eta. \tag{12}$$

For $k \ll k_*$, the solution is $G(k) = G_N(k_*/k)^\eta$. Kumar [17] showed from QFT first principles that $\eta = 1$ is the unique marginal value: it is the boundary between corrections that die off ($\eta < 1$) and corrections that grow without bound ($\eta > 1$). Within the $\tau$ framework, this value is independently reproduced by the DPI plus extensivity of de Sitter entanglement entropy (see Sec. IX for a sketch of the argument; the complete proof requires the perturbative RG analysis of Paper IV [52]). In position space, $\eta = 1$ produces the unique scale-invariant, rotationally symmetric deformation of the Newtonian potential:

$$G(r) = G_N\left[1 + \frac{2k_* r}{\pi}\right], \quad r \gg 1/k_*, \tag{13}$$

with the coefficient $2/\pi$ being universal and regulator-independent.

The modified potential, circular velocity, and effective entropy production are:

$$\Phi(r) = -\frac{G_N M}{r} + \frac{2G_N M k_*}{\pi}\ln\left(\frac{r}{r_0}\right), \tag{14}$$

$$v^2(r) = \frac{G_N M}{r} + \frac{2G_N M k_*}{\pi}, \tag{15}$$

$$\Sigma_{\rm grav}(r) = \frac{r_s}{r} + \frac{2k_* r_s}{\pi}\ln\left(\frac{r}{r_0}\right), \tag{16}$$

where $r_s = 2G_N M/c^2$ is the Schwarzschild radius and $r_0$ is a reference scale. Equation (15) yields a flat rotation curve at $r \gg 1/k_*$:

$$V_0^2 = \frac{2G_N M_{\rm bar}\, k_*}{\pi}, \tag{17}$$

which is constant.

The logical chain connecting established results is:

RG flow is a quantum channel [14]

$\xrightarrow{\text{DPI}}$ QRE monotonicity under RG

$\xrightarrow{\text{Dorau-Much}}$ QRE $\to$ semiclassical Einstein eqns. [16]

$\xrightarrow{\text{Kumar}}$ Running $G$ with $\eta = 1 \to$ flat rotation curves [17]

### C. SPARC galaxies and the universal crossover scale

Gubitosi, Piattella, and Casarini [20] validated the running-$G$ framework (termed RGGR) against 100 galaxies from the SPARC database [21], finding it competitive with NFW dark matter halos (RGGR preferred for 47 galaxies, NFW for 40, inconclusive for 13 by the BIC criterion).

TABLE I. Crossover scale $k_*$ for three SPARC galaxies from Ref. [17]. The near-universality of $k_*$ across a factor of $\sim$3 in baryonic mass is a specific prediction of the framework.

| Galaxy | $M_{\rm bar}$ [$M_\odot$] | $V_0$ [km/s] | $k_*$ [kpc$^{-1}$] |
|---|---|---|---|
| S:700624 | $1.01 \times 10^{12}$ | $271 \pm 3$ | $(2.66 \pm 0.05)\times 10^{-2}$ |
| S:700916 | $5.85 \times 10^{11}$ | $210 \pm 2$ | $(2.76 \pm 0.06)\times 10^{-2}$ |
| S:705253 | $3.55 \times 10^{11}$ | $159 \pm 2$ | $(2.60 \pm 0.06)\times 10^{-2}$ |

Kumar [17] fitted three SPARC galaxies and found a consistent crossover scale (Table I):

The universal $k_* \approx 2.7 \times 10^{-2}$ kpc$^{-1}$ (crossover radius $r_c = 1/k_* \approx 37$ kpc) is a specific prediction absent from NFW halo models, which require two parameters ($M_{200}$, concentration $c$) per galaxy.

### D. Baryonic Tully–Fisher relation

From Eq. (17), for universal $k_*$:

$$M_{\rm bar} = \frac{\pi}{2G_N k_*} V_0^2. \tag{18}$$

Since $k_*$ is approximately constant across galaxies, this is a linear relation $M_{\rm bar} \propto V_0^2$. The observed baryonic Tully–Fisher relation $M_{\rm bar} \propto V_0^4$ [6] is recovered in the deep MOND limit where the Verlinde contribution dominates (Sec. IV). The running-$G$ contribution gives the correct intermediate-radius behavior, while the Verlinde contribution governs the asymptotic regime where $v^4 = a_0 G_N M_{\rm bar}$.

### E. Radial acceleration relation

The RAR [5] relates observed acceleration $g_{\rm obs}$ to baryonic acceleration $g_{\rm bar}$ via

$$g_{\rm obs} = \frac{g_{\rm bar}}{1 - e^{-\sqrt{g_{\rm bar}/g_\dagger}}}, \tag{19}$$

with $g_\dagger = a_0 = (1.20 \pm 0.02) \times 10^{-10}$ m/s$^2$. In the $\tau$ framework, the running-$G$ correction at the galactic scale produces

$$g_{\rm obs} = g_{\rm bar} + \frac{2G_N M_{\rm bar}\, k_*}{\pi r^2}, \tag{20}$$

which reproduces the deep-MOND behavior $g_{\rm obs} \sim \sqrt{a_0\, g_{\rm bar}}$ when $k_*$ is related to $a_0$ through the cosmological connection (Sec. IV).

### F. Dark matter density profile

The apparent dark matter density from Eq. (16) is

$$\rho_{\rm DM}(r) = \frac{1}{4\pi G_N}\frac{k_* V_0^2}{r^2} \propto \frac{1}{r^2}, \tag{21}$$

which gives an enclosed mass $M_{\rm DM}(r) \propto r$. This is a $1/r^2$ density profile, steeper than isothermal ($1/r^2$, coincidentally the same) but distinct from the NFW profile ($1/r$ at small $r$, $1/r^3$ at large $r$). The key prediction is the *absence* of a break: the $1/r^2$ profile extends to arbitrarily large $r$ without the $1/r^3$ falloff characteristic of NFW halos. This is testable with weak lensing data extending to the virial radius and beyond. The KiDS-1000 weak lensing analysis [22] already extends the RAR by two decades in acceleration and is consistent with the absence of a break.

### G. Dwarf spheroidal galaxies

Ghari and Haghi [23] compared Verlinde's emergent gravity to MOND for 23 Milky Way dwarf spheroidal satellites and found Verlinde favored over MOND at $5.2\sigma$. Since the Verlinde contribution is a subset of the $\tau$ framework's $\Sigma_{\rm DM}$ (the volume-law entanglement component), this result supports the information-theoretic interpretation.

### H. Solar system constraints

At solar system scales ($r \ll r_c$), the running-$G$ correction is negligible: $G(r)/G_N - 1 \sim 2k_* r/\pi \sim 2\times 10^{-14}$ at $r = 1$ AU, well below current constraints $|G_{\rm eff}/G_N - 1| < 10^{-5}$ [24].

## IV. THE MOND SCALE FROM THE CROOKS THEOREM

### A. The $a_0$–cosmology coincidence

The empirical MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s$^2$ is numerically close to $cH_0$:

$$\frac{cH_0}{a_0} \approx 5.7, \tag{22}$$

with $H_0 = 70$ km/s/Mpc. More precisely, $a_0 \approx cH_0/(2\pi) \approx cH_0/6$ (these differ by $\sim 5\%$). The coincidence has been noted by Milgrom [25] and is a key input to Verlinde's derivation [9], where $a_0 = cH_0/6$.

### B. KMS–Crooks argument

We provide a thermodynamic argument for this connection. The de Sitter vacuum satisfies the KMS condition with modular time periodicity

$$\beta_{\rm dS} = \frac{2\pi}{H_0}. \tag{23}$$

The spatial thermal wavelength is

$$\lambda_{\rm dS} = c\,\beta_{\rm dS} = \frac{2\pi c}{H_0}, \tag{24}$$

and the associated acceleration scale is

$$a_0 = \frac{c^2}{\lambda_{\rm dS}} = \frac{cH_0}{2\pi} \approx 1.08\times 10^{-10}\ {\rm m/s^2}. \tag{25}$$

The Crooks fluctuation theorem [26, 27] gives the ratio of forward to reverse transition probabilities as $P_F(\Sigma)/P_R(-\Sigma) = e^{\Sigma}$. For a gravitational process, the entropy production per degree of freedom is

$$\Sigma = \frac{k_{\rm B}T_U}{k_{\rm B}T_{\rm dS}} = \frac{a}{cH_0}, \tag{26}$$

where $T_U = \hbar a/(2\pi c k_{\rm B})$ is the Unruh temperature at acceleration $a$ and $T_{\rm dS} = \hbar H_0/(2\pi k_{\rm B})$.

Three regimes emerge:

- $a \gg a_0$ ($\Sigma \gg 1$): Gravitational processes are irreversible. The forward process dominates by a factor $e^{\Sigma} \gg 1$. The time arrow is strong. This is the Newtonian regime.
- $a \sim a_0$ ($\Sigma \sim 1$): Forward and reverse processes have comparable probabilities ($\sim 63\%$ vs. $\sim 37\%$). Irreversibility is *marginal*. This is the MOND transition.
- $a \ll a_0$ ($\Sigma \ll 1$): Processes are nearly reversible. The time arrow weakens. This is the deep MOND regime, where the de Sitter thermal bath dominates.

The MOND transition is therefore the transition from strong to marginal irreversibility in the gravitational channel, as measured by the Crooks entropy production [28].

### C. Crooks interpolation function

The exponential form of the Crooks ratio suggests a natural MOND interpolation function. Writing $x = g_{\rm bar}/a_0$, we propose

$$\mu(x) = 1 - e^{-x}. \tag{27}$$

The effective acceleration is $g_{\rm obs} = g_{\rm bar}/\mu(x)$. Properties of Eq. (27):

$$\begin{aligned} x \gg 1: &\quad \mu \to 1 \quad \text{(Newtonian)},\\ x \ll 1: &\quad \mu \approx x \quad \text{(deep MOND)}. \end{aligned}$$

This is the same asymptotic behavior as the standard MOND interpolation functions. Numerically, for representative values:

$$\begin{aligned} x = 0.1: &\quad \mu_{\text{Crooks}} = 0.095, \quad \mu_{\text{std}} = 0.091, \\ x = 1.0: &\quad \mu_{\text{Crooks}} = 0.632, \quad \mu_{\text{std}} = 0.500, \\ x = 3.0: &\quad \mu_{\text{Crooks}} = 0.950, \quad \mu_{\text{std}} = 0.750. \end{aligned}$$

The Crooks form transitions faster from MOND to Newton than the standard $\mu(x) = x/(1+x)$; whether this is preferred by the data requires systematic fitting to SPARC galaxies, which we leave to future work.

### D. Limitations of the $a_0$ argument

The identification of $a_0 = cH_0/(2\pi)$ should be classified as a *dimensional argument with physical content*, not a rigorous derivation. The scale $cH_0$ follows inevitably from combining the speed of light with the Hubble rate. The factor $2\pi$ is motivated by the KMS periodicity of the de Sitter thermal state but is not uniquely derived from the Crooks theorem. Verlinde's factor of 6 arises from a different geometric argument (the volume-to-area ratio of a sphere combined with the quadratic elastic energy). Since $2\pi \approx 6.28 \approx 6$ to within 5%, these predictions are observationally indistinguishable with current data.

The situation is analogous to the Chandrasekhar mass: $M_{\text{Ch}} \sim (\hbar c/G)^{3/2}/m_p^2$ follows from dimensional analysis, but the coefficient 0.77 requires solving the Lane–Emden equation. Similarly, the precise $O(1)$ prefactor in $a_0$ awaits a more detailed calculation of the Crooks entropy production for test particles in de Sitter spacetime, using the formalism of Ref. [28].

## V. THE BULLET CLUSTER

The Bullet Cluster (1E 0657-558) [3] poses the sharpest challenge to any theory that explains dark matter through modified gravity or emergent effects. The lensing mass peaks are offset from the X-ray gas (which contains $\sim 85\%$ of the baryonic mass) by $\sim 150$ kpc, coinciding instead with the galaxy distribution.

### A. The challenge for modified gravity

For a theory where the gravitational modification is proportional to the local baryonic mass (as in running $G$), the gas dominates the total gravitational signal. Running $G$ cannot by itself produce a lensing peak offset from the gas. This is a generic problem shared by all modified gravity theories that have been tested against the Bullet Cluster [29], with the exception of MOG/STVG [30], whose additional vector field provides the needed offset.

### B. Entanglement rigidity argument

We offer a qualitative argument based on the dynamical timescales. The volume-law entanglement pattern that produces the apparent dark matter (the Verlinde component of $\Sigma_{\text{DM}}$) was established over the equilibration timescale $t_{\text{eq}} \sim H_0^{-1} \sim 10^{10}$ yr. The cluster collision timescale is $t_{\text{coll}} \sim 10^8$ yr. The ratio

$$\frac{t_{\text{eq}}}{t_{\text{coll}}} \sim 100 \tag{28}$$

suggests that the entanglement pattern responds sluggishly to the rapid collision. During the merger, the gas is ram-pressure stripped and decelerates, but the entanglement displacement pattern—associated with the pre-collision mass distribution—cannot fully re-equilibrate. If $\sim 95\%$ of the entanglement-based $\Sigma_{\text{DM}}$ is "frozen" during the collision, the lensing signal would partially track the pre-collision mass distribution, which is centered on the galaxy positions.

This argument is qualitative and speculative. It requires a theory of entanglement equilibration timescales in de Sitter space that does not currently exist. We note that the Bullet Cluster is also problematic for ΛCDM: the inferred collision velocity ($\sim 3000$ km/s) is statistically improbable [31], and the Abell 520 cluster shows the *opposite* pattern (a lensing peak without galaxies), which CDM also struggles to explain [32].

## VI. COSMOLOGICAL SCALE: DARK ENERGY

At cosmological scales, the entropy production from the de Sitter vacuum gives

$$\Sigma_{\text{dS}}(r) = -\ln\left(1 - \frac{H_0^2 r^2}{c^2}\right), \tag{29}$$

which diverges at the Hubble horizon $r_H = c/H_0$, corresponding to $\tau \to 1$: complete information loss at the cosmological horizon. This is the ultimate gravitational $\tau$.

Padmanabhan's CosMIn framework [35] establishes that finite cosmic information requires $\Lambda > 0$. In the $\tau$ framework, this translates to: a finite total information loss ($\int \Sigma\, dt < \infty$) over the history of the universe requires a cosmic information horizon where $\tau = 1$. This is precisely the de Sitter horizon set by $\Lambda$.

The connection between the cosmological constant and the de Sitter thermal state provides the bridge to dark matter: the *same* vacuum entanglement that produces $\Lambda > 0$ (through the volume-law contribution) also produces the apparent dark matter at galactic scales (through its displacement by baryonic matter). In the $\Sigma$ language:

$$\Sigma_{\text{total}}(r) \approx \underbrace{\frac{r_s}{r}}_{\text{Newtonian}} + \underbrace{\alpha \ln\left(\frac{r}{r_c}\right)}_{\text{running } G} + \underbrace{\beta \frac{r}{L_{\text{dS}}}}_{\text{de Sitter}}, \tag{30}$$

where $L_{\rm dS} = c/H_0$. Each term dominates in a different regime: the Newtonian term near the mass source, the running-$G$ term at galactic scales, and the de Sitter term at cosmological scales. All three emerge as limits of the single quantum relative entropy $\Sigma = D(\rho_{\rm spacetime}\|\rho_{\rm matter})$ [13, 34].

The DESI DR2 results showing $w(z) \neq -1$ at $\sim 4\sigma$ [36] are naturally compatible with the $\tau$ framework, since a scale-dependent $\Sigma_{\rm grav}$ generically produces evolving dark energy. We note this as a qualitative consistency, not a prediction.

## VII. THE CMB GAP

The $\tau$ framework, as currently formulated, *cannot* explain the CMB acoustic peaks without introducing additional degrees of freedom. This section presents the argument in full, because the inability to solve this problem is as informative as the successes.

### A. The quantitative failure

The CMB third acoustic peak requires a pressureless ($c_s^2 = 0$), photon-decoupled component with $\Omega h^2 \approx 0.12$. Without such a component, the predicted first peak position shifts from $\ell \approx 220$ to $\ell \approx 170$ (a 23% error, $> 100\sigma$), the third-to-second peak ratio drops from 0.97 to $\sim 0.55$ (a 43% error, $> 20\sigma$), and the matter-radiation equality redshift shifts from $z_{\rm eq} \approx 3400$ to $z_{\rm eq} \approx 530$ (a factor 6.4 discrepancy) [2, 37]. These are catastrophic failures, not marginal discrepancies.

### B. Why running $G$ is irrelevant at CMB scales

The IR running formula $G(k) = G_N[1 + k_*/k]$ with $k_* \sim 10^{-2}$ kpc$^{-1}$ gives $G/G_N \sim 2700$ at CMB scales ($k_{\rm CMB} \sim 10^{-2}$ Mpc$^{-1}$), which is physically absurd. The running is a late-time, galactic-scale effect that does not apply at $z \sim 1100$. Constraints from Planck require $|G(z{\sim}1100)/G_0 - 1| < 0.05$ at 95% CL.

### C. The fundamental obstruction

The core problem is that any perturbation mode derived from a local, Lorentz-invariant quantum field theory on an FRW background has a sound speed $c_s > 0$. The quantum relative entropy $D(\rho_{\rm spacetime}\|\rho_{\rm matter})$ is Lorentz invariant, so its perturbations propagate at the speed of light. But the CMB requires $c_s = 0$—a non-propagating mode.

The only known ways to achieve $c_s = 0$ are: (i) breaking Lorentz invariance in the gravity sector (as in AeST's aether [8] and the Khronon field [33]), (ii) introducing massive non-relativistic particles (CDM), or (iii) imposing a non-propagating constraint via a Lagrange multiplier (as in Bianconi's $G$-field [34]).

TABLE II. Structural parallel between the $\tau$ framework and relativistic MOND theories. The observer's 4-velocity $u^\mu$ plays the same structural role as the aether field $A^\mu$.

| | **$\tau$ framework** | **AeST/Khronon** |
|---|---|---|
| Scalar field | $\Sigma$ (entropy prod.) | $\tau$ (Khronon) |
| Background | $\Sigma = 0$ (comoving) | $Q = 1$ (comoving) |
| Observer/aether | $u^\mu$ (observer) | $A^\mu$ (aether) |
| MOND limit | Running $G$ | $J(Y)$ function |
| CMB behavior | Running $\mu$, dust-like at $z > 100$ | GDM: $w{\sim}0$, $c_s{\sim}0$ |

### D. Structural parallel with AeST and the Modular Khronon

The parallel between the $\tau$ framework and the Khronon/AeST theories is notable (Table II):

In the Khronon theory [33], the DBI-inspired kinetic function $K(Q) = \mu^2(Q-1)^2$ has a minimum at $Q = 1$ (unit timelike gradient), producing a mode with $\omega = 0$ (non-propagating) on the Minkowski background. This is the mechanism that yields $c_s = 0$ and hence CDM-like behavior.

In the $\tau$ framework, the observer's 4-velocity $u^\mu$ plays the role of the Khronon gradient $\nabla^\mu\tau$. If $u^\mu$ is promoted from an external input to a dynamical field—constrained by $g_{\mu\nu}u^\mu u^\nu = -1$ (unit timelike)—the resulting theory would naturally contain: (i) a preferred foliation (breaking Lorentz invariance in the gravity sector), (ii) a scalar mode with $c_s = 0$ from the unit-timelike constraint, and (iii) a connection to the AeST/Khronon field content.

This "observer-as-aether" identification is a natural route to closing the CMB gap. It changes the framework fundamentally: the observer is no longer external but becomes part of the dynamics. If successful, dark matter at cosmological scales would be *the energetic cost of maintaining a time direction in the universe*. Paper IV develops this idea: the modular Khronon $\delta T_{\rm mod} = -2\Phi/H$ emerges from QRE perturbation theory and has $c_s = 0$ in the matter era, providing exactly the CDM-like perturbation behavior needed for the CMB [52].

### E. The CMB gap in context

The CMB gap is shared by *all* modified gravity approaches. MOND [7] has no CMB prediction without a relativistic completion. Verlinde [9] has no perturbation theory. The only existing modified gravity theory that fits both galactic rotation curves and the CMB is AeST [8], and it achieves this by introducing precisely the field content (vector + scalar) that our analysis iden-

tifies as structurally necessary. The $\tau$ framework's contribution is to identify *why* this field content is needed: to break Lorentz invariance in a way that produces a $c_s = 0$ mode.

### F. Resolution via running Khronon

The CMB gap identified above has a candidate resolution within the $\tau$ framework once the Khronon coupling $\mu$ is promoted from a free parameter to a dimensionally-determined quantity.

*Dimensional uniqueness of $\mu_0$.*—The Khronon kinetic function $K(Q) = \mu^2(Q-1)^2$ [33] requires a coupling $\mu$ with dimensions of inverse length. In de Sitter space, the only available scales are $H_0$, $c$, $\hbar$, and $k_{\rm B}$. The unique combination with dimensions of inverse length that is independent of $\hbar$ (as befits a classical coupling) is

$$\mu_0 = \frac{H_0}{c} = \frac{2\pi k_{\rm B} T_{\rm dS}}{\hbar c}, \tag{31}$$

where $T_{\rm dS} = \hbar H_0/(2\pi k_{\rm B})$ is the de Sitter temperature. This is not an assumption but a consequence of dimensional analysis: the same thermal scale that determines $a_0 = cH_0/(2\pi)$ also determines $\mu_0$.

*Running of $\mu$.*—The anomalous dimension of $\mu$ follows from the same DPI + extensivity argument that fixes $\eta_G = 1$ for the gravitational coupling (Sec. III):

$$\eta_\mu = 1, \qquad \mu(k) = \mu_0\left(\frac{k_*}{k}\right), \tag{32}$$

because the Khronon field couples to gravity through the same renormalization group flow. The DPI forbids $\eta_\mu > 1$ (would violate the de Sitter entropy bound on the Khronon sector), while $\eta_\mu < 1$ makes the Khronon IR-irrelevant, contradicting its required role at CMB scales. $\eta_\mu = 1$ is the unique marginal value.

*Dust-like behavior at CMB scales.*—On an FRW background, the Khronon gradient satisfies $Q = 1$ (comoving frame), and the background coupling runs as

$$\mu_{\rm bg}(a) = \frac{H(a)}{c}, \tag{33}$$

where $H(a)$ is the Hubble parameter at scale factor $a$. The effective equation-of-state parameter for the Khronon perturbations is [33]

$$\tilde{w}_0 = \frac{\mu_{\rm bg}^2}{k_{\rm phys}^2}, \tag{34}$$

where $k_{\rm phys}$ is the physical wavenumber of the perturbation mode. At recombination ($z \approx 1100$), $H(z_{\rm rec}) \approx 2.3 \times 10^{-14}$ s$^{-1}$ and the first acoustic peak corresponds to $k_{\rm phys} \sim 0.01$ Mpc$^{-1}$, giving

$$\tilde{w}_0(z \approx 1100) \sim 6 \times 10^{-11} \ll 1. \tag{35}$$

This is effectively pressureless: the Khronon perturbations behave as dust at CMB scales, precisely the $c_s = 0$ mode identified as structurally necessary in Sec. VII.

*Unification.*—The de Sitter temperature $T_{\rm dS}$ now serves as the single origin of both galactic and cosmological dark-matter phenomenology:

$$T_{\rm dS} \longrightarrow \begin{cases} a_0 = cH_0/(2\pi) & \text{(galactic: MOND scale)}, \\ \mu_0 = H_0/c & \text{(CMB: Khronon coupling)}. \end{cases} \tag{36}$$

At galactic scales, the running of $G$ with $\eta_G = 1$ provides MOND-like rotation curves (Sec. III). At CMB scales, the running of $\mu$ with $\eta_\mu = 1$ provides a candidate for the pressureless, non-propagating mode needed for the acoustic peaks. Both originate from the same de Sitter vacuum entanglement. Paper IV [52] develops the full cosmological perturbation theory; here we note only that the mechanism exists and produces the correct qualitative behavior.

*Status: experimentally unverified.*—The running Khronon resolution has not been tested against the full Planck CMB power spectrum. The qualitative argument (dust-like $\tilde{w}_0 \ll 1$) is encouraging but does not constitute a quantitative fit. Until the numerical computation of Paper IV reproduces the observed $C_\ell^{TT}$ spectrum to within Planck error bars, this resolution should be regarded as a conjecture.

*Limitations.*—The $\Omega_{\rm DM}h^2$ prefactor—the precise numerical value that Planck measures as $0.120 \pm 0.001$—is *not* determined by the dimensional argument alone. It depends on the full nonlinear dynamics of the Khronon field on an FRW background, which requires numerical integration beyond the scope of this paper. We predict the dark matter density to order of magnitude (from $T_{\rm dS}$ and $\mu_0$); the exact prefactor requires the numerical computation presented in Paper IV.

## VIII. TESTABLE PREDICTIONS

The $\tau$ framework makes five testable predictions distinguishable from ΛCDM and standard MOND:

*1. DM density profile $\propto 1/r^2$ without NFW break.*—The running-$G$ correction produces an enclosed mass $M_{\rm DM}(r) \propto r$ at large $r$, without the $M_{\rm DM} \to$ const behavior of NFW halos. For a Milky Way–mass galaxy ($M_{\rm bar} \approx 5 \times 10^{10}\ M_\odot$), the predicted dark matter density at $r = 100$ kpc is $\rho_{\rm DM} \approx 3 \times 10^{-4}\ M_\odot/{\rm pc}^3$, declining as $r^{-2}$ with no steepening. Weak lensing profiles extending beyond the virial radius should show no turnover. The BIG-SPARC survey [38] with $\sim 4000$ galaxies can test this.

*2. Universal crossover scale and running $G$.*—The crossover momentum $k_* \approx 2.7 \times 10^{-2}$ kpc$^{-1}$ (crossover radius $r_c \approx 37$ kpc) should be independent of galaxy mass (within observational scatter). Concretely, $G(r)/G_N = 1 + 2k_* r/\pi$ predicts: $G/G_N \approx 1.02$ at $r = 1$ kpc, $G/G_N \approx 1.17$ at $r = 10$ kpc, and

$G/G_N \approx 2.72$ at $r = 100$ kpc. A systematic variation $k_*(M_{\rm bar})$ would indicate environment-dependent running rather than a universal fundamental scale.

*3. Interpolation function* $\mu(x) = 1 - e^{-x}$.—The Crooks-inspired interpolation transitions faster from MOND to Newtonian than the standard $\mu(x) = x/(1+x)$. At $x = 1$, $\mu_{\rm Crooks} = 0.63$ vs. $\mu_{\rm std} = 0.50$. This is distinguishable with high-quality rotation curves near the transition radius.

*4. Observer-dependent* $a_0$.—Since $\Sigma$ is observer-dependent [28], the MOND acceleration scale should depend on the observer's worldline. Comoving observers see the standard $a_0 \approx cH_0/(2\pi) \approx 1.08 \times 10^{-10}$ m/s$^2$, while observers near massive objects (galaxy clusters, black holes) should see a locally modified $a_0$ due to gravitational redshift. Specifically, at gravitational potential $\Phi$:

$$a_0^{\rm local} = a_0 \sqrt{-g_{00}} \approx a_0 \left(1 + \frac{\Phi}{c^2}\right). \tag{37}$$

For galaxies in a cluster with $\Phi/c^2 \sim -10^{-5}$, the predicted shift is $\delta a_0/a_0 \sim 10^{-5}$—small but in principle measurable through precision RAR studies of cluster satellite galaxies compared to field galaxies.

*5. Lensing vs. kinematic DM mass may differ.*—If $\Sigma_{\rm DM}$ has both running-$G$ and Verlinde components with different radial profiles, the "dark matter mass" inferred from lensing (which probes $g_{00} + g_{rr}$) may differ from that inferred from kinematics (which probes $g_{00}$ alone). In GR, these are identical for static metrics; in the $\tau$ framework, the logarithmic correction modifies both $g_{00}$ and $g_{rr}$ equally, preserving the GR prediction. However, at the transition between the running-$G$ and Verlinde regimes, deviations are possible.

## IX. DISCUSSION

### A. What is genuinely new

Six elements distinguish this work from previous approaches to the dark matter problem:

(i) The identification of "dark matter" with observer-dependent entropy production $\Sigma_{\rm DM} = \Sigma_{\rm cl} - \Sigma_{\rm q}$, which is guaranteed non-negative by the DPI (Theorem 1). This is not a model assumption but a mathematical consequence of the framework.

(ii) The logical chain connecting the Casini–Huerta entropic $c$-theorem [14, 15], the Dorau–Much QRE-to-Einstein link [16], and the Kumar running $G$ [17] into a coherent information-theoretic narrative for galactic rotation curves. Each step is established; the synthesis is new.

(iii) The identification of $a_0 \approx cH_0/(2\pi)$ with the KMS thermal periodicity of de Sitter space through the Crooks theorem, interpreting the MOND transition as marginal irreversibility (Sec. IV; note that the $O(1)$ prefactor is a dimensional argument, not a rigorous derivation).

(iv) The Crooks interpolation function $\mu(x) = 1 - e^{-x}$, which is a specific, falsifiable prediction.

(v) The identification of the observer's 4-velocity with the aether/Khronon field, providing a concrete bridge between the $\tau$ framework and relativistic MOND theories.

(vi) The identification of the Khronon coupling $\mu_0 = H_0/c$ from dimensional uniqueness, and $\eta_\mu = 1$ from DPI + extensivity, yielding dust-like behavior at CMB scales without introducing new free parameters (Sec. VII F). This CMB resolution is not yet verified against the full Planck power spectrum; the quantitative test requires Paper IV.

### B. What is not claimed

We do *not* claim: (i) that the CMB acoustic peaks are quantitatively fitted—the running Khronon coupling $\mu_0 = H_0/c$ with $\eta_\mu = 1$ provides CDM-like behavior at CMB scales (Sec. VII F), but the exact $\Omega_{\rm DM}h^2$ prefactor remains undetermined without the full numerical computation of Paper IV; (ii) that the Bullet Cluster is quantitatively fitted (only a qualitative argument is given); or (iii) superiority over ΛCDM at all scales. We *do* note that $\eta = 1$ is derived, not assumed: Kumar [17] identified $\eta = 1$ as the unique marginal value from QFT first principles. Within the $\tau$ framework, this is independently reproduced by the DPI applied to the gravitational RG flow, combined with the extensivity of de Sitter entanglement entropy (volume-law scaling at cosmological scales [9]), which uniquely selects $\eta = 1$ in $3+1$ dimensions. The argument is: $\eta > 1$ violates the de Sitter entropy bound; $\eta < 1$ contradicts extensivity (the correction becomes IR-irrelevant); $\eta = 1$ is the unique marginal value producing the logarithmic potential. In general $d$ dimensions, $\eta = d-3$. The same DPI + extensivity argument fixes $\eta_\mu = 1$ for the Khronon coupling (Sec. VII F), providing the mechanism—though not yet the precise numerical coefficient—for CMB-scale dark matter behavior.

### C. The three-term Σ as a unified equation

Equation (30) approximates the single quantum relative entropy $\Sigma = D(\rho_{\rm spacetime} \| \rho_{\rm matter})$ in three regimes. In the vision of the four-paper series [11, 13, 39], different boundary conditions on this one QRE give: Paper I ($\Sigma = 0$): flat spacetime, perfect recovery; Paper II ($\Sigma = r_s/r$): Schwarzschild, the time arrow of gravity; Paper III ($\Sigma = r_s/r + \alpha \ln(r/r_c)$): galactic scales, dark matter; Paper IV ($\Sigma$ with cosmological boundary conditions): dark energy and the CMB. These are not four separate theories but four limits of one equation, analogous to $F = ma$ producing planetary orbits, spring oscillations, and fluid dynamics depending on boundary conditions.

### D. The CMB gap as a signpost

The failure at CMB scales is not a deficiency to be papered over but a signpost. It points to a specific mathematical program: determine whether the constrained minimization of $D(\rho_{\text{spacetime}}\|\rho_{\text{matter}})$ produces Lagrange multiplier fields with $c_s = 0$. If it does, the $\tau$ framework would derive CDM-like behavior from quantum information theory. If it does not, then CDM particles are a necessary ingredient that the framework must incorporate rather than explain. Either outcome would clarify the scope of the framework.

The running Khronon analysis of Sec. VII F suggests an affirmative answer: the constrained observer dynamics produces a dust-like mode at CMB scales, via the running coupling $\mu_0 = H_0/c$. Whether this survives a full numerical comparison with the Planck power spectrum remains to be determined (Paper IV).

### E. Division of labor: $G$ and $\mu$

The $\tau$ framework now exhibits a clear division of labor between two running couplings, both originating from the de Sitter vacuum and both with anomalous dimension $\eta = 1$:

*Running* $G$ governs galactic scales. The infrared running $G(k) = G_N(1 + k_*/k)$ produces the logarithmic correction to the Newtonian potential, flat rotation curves, and the radial acceleration relation. The crossover scale $k_* \sim 10^{-2}$ kpc$^{-1}$ is set by $a_0 = cH_0/(2\pi)$. This is a modification of the *strength* of gravity.

*Running* $\mu$ governs cosmological scales. The Khronon coupling $\mu(k) = \mu_0(k_*/k)$ with $\mu_0 = H_0/c$ determines the dynamics of the observer field. At CMB scales, $\mu_{\text{bg}}(a) = H(a)/c$ produces a pressureless mode ($\tilde{w}_0 \ll 1$) that mimics cold dark matter perturbations. This is a modification of the *observer dynamics*—the energetic cost of maintaining a time direction.

The two effects are complementary, not redundant: running $G$ is a late-time, small-scale effect that is irrelevant at $z \sim 1100$ (Sec. VII), while running $\mu$ is a large-scale effect whose galactic contribution is subdominant to the running-$G$ correction. Together, they cover the full range from kiloparsec to gigaparsec scales with a single origin: the de Sitter temperature $T_{\text{dS}}$.

## ACKNOWLEDGMENTS

The author thanks M. M. Wilde for valuable feedback on the $\tau$ framework, and A. J. Parzygnat and F. Buscemi for foundational work on retrodiction.

[1] V. C. Rubin, W. K. Ford, Jr., and N. Thonnard, "Rotational properties of 21 Sc galaxies with a large range of luminosities and radii, from NGC 4605 ($R = 4$ kpc) to UGC 2885 ($R = 122$ kpc)," Astrophys. J. **238**, 471 (1980).

[2] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020); arXiv:1807.06209.

[3] D. Clowe, M. Bradač, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones, and D. Zaritsky, "A direct empirical proof of the existence of dark matter," Astrophys. J. Lett. **648**, L109 (2006); astro-ph/0608407.

[4] M. Schumann, "Direct detection of WIMP dark matter: Concepts and status," J. Phys. G **46**, 103003 (2019); arXiv:1903.03026.

[5] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial acceleration relation in rotationally supported galaxies," Phys. Rev. Lett. **117**, 201101 (2016); arXiv:1609.05917.

[6] S. S. McGaugh, J. M. Schombert, G. D. Bothun, and W. J. G. de Blok, "The baryonic Tully–Fisher relation," Astrophys. J. Lett. **533**, L99 (2000); astro-ph/0003001.

[7] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis," Astrophys. J. **270**, 365 (1983).

[8] C. Skordis and T. Zlošnik, "New relativistic theory for modified Newtonian dynamics," Phys. Rev. Lett. **127**, 161302 (2021); arXiv:2007.00082.

[9] E. P. Verlinde, "Emergent gravity and the dark universe," SciPost Phys. **2**, 016 (2017); arXiv:1611.02269.

[10] S. Ettori, V. Ghirardini, D. Eckert, E. Pointecouteau, F. Gastaldello, M. Sereno, M. Gaspari, S. Ghizzardi, M. Roncarelli, and M. Rossetti, "Hydrostatic mass profiles in X-COP galaxy clusters," Astron. Astrophys. **621**, A39 (2019); arXiv:1805.00035.

[11] S.-K. Huang, "Observer-dependent temporal asymmetry: From quantum erasure to gravitational time" (Paper V), in preparation (2026).

[12] M. Junge, R. Renner, D. Sutter, M. M. Wilde, and A. Winter, "Universal recovery maps and approximate sufficiency of quantum relative entropy," Ann. Henri Poincaré **19**, 2955 (2018).

[13] S.-K. Huang, "Gravitational temporal asymmetry from information loss" (Paper II), in preparation (2026).

[14] H. Casini, E. Teste, and G. Torroba, "Relative entropy and the RG flow," J. High Energy Phys. **03**, 089 (2017); arXiv:1611.00016.

[15] H. Casini and M. Huerta, "On the RG running of the entanglement entropy of a circle," Phys. Rev. D **85**, 125016 (2012); arXiv:1202.5650.

[16] V. Dorau and A. Much, "From quantum relative entropy to the semiclassical Einstein equations," Phys. Rev. Lett. **136**, 091602 (2026); arXiv:2510.24491.

[17] N. Kumar, "Marginal IR running of gravity as a natural explanation for dark matter," Phys. Lett. B **871**, 140008 (2025); arXiv:2509.05246.

[18] M. Reuter and H. Weyer, "Running Newton constant, improved gravitational actions, and galaxy rotation curves," Phys. Rev. D **70**, 124028 (2004); hep-

th/0410117.

[19] J. F. Donoghue, "General relativity as an effective field theory: The leading quantum corrections," Phys. Rev. D **50**, 3874 (1994); gr-qc/9405057.

[20] G. Gubitosi, O. F. Piattella, and L. Casarini, "Phenomenology of renormalization group improved gravity from the kinematics of SPARC galaxies," Phys. Rev. D **110**, 124014 (2024); arXiv:2403.00531.

[21] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves," Astron. J. **152**, 157 (2016); arXiv:1606.09251.

[22] M. M. Brouwer *et al.*, "The radial acceleration relation in GAMA/KiDS: Bridging the gap between theory and observations," Astron. Astrophys. **650**, A113 (2021); arXiv:2106.11677.

[23] A. Ghari and H. Haghi, "Verlinde's emergent gravity vs. MOND in dwarf spheroidal galaxies," arXiv:2601.01715 (2026).

[24] C. M. Will, "The confrontation between general relativity and experiment," Living Rev. Relativ. **17**, 4 (2014); arXiv:1403.7377.

[25] M. Milgrom, "The $a_0$–cosmology connection in MOND," arXiv:2001.09729 (2020).

[26] G. E. Crooks, "Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences," Phys. Rev. E **60**, 2721 (1999); cond-mat/9901352.

[27] H. Kwon and M. S. Kim, "Fluctuation theorems for a quantum channel," Phys. Rev. X **9**, 031029 (2019).

[28] M. L. W. Basso, J. Maziero, and L. C. Celeri, "Quantum detailed fluctuation theorem in curved spacetimes: The observer dependent nature of entropy production," Phys. Rev. Lett. **134**, 050406 (2025); arXiv:2405.03902.

[29] B. Famaey and S. S. McGaugh, "Modified Newtonian dynamics (MOND): Observational phenomenology and relativistic extensions," Living Rev. Relativ. **15**, 10 (2012); arXiv:1112.3960.

[30] J. R. Brownstein and J. W. Moffat, "The Bullet Cluster 1E0657-558 evidence shows modified gravity in the absence of dark matter," Mon. Not. R. Astron. Soc. **382**, 29 (2007); astro-ph/0702146.

[31] J. Lee and E. Komatsu, "Bullet Cluster: A challenge to ΛCDM cosmology," Astrophys. J. **718**, 60 (2010); arXiv:1003.0939.

[32] M. J. Jee *et al.*, "Discovery of a ringlike dark matter structure in the core of the galaxy cluster Cl 0024+17," Astrophys. J. **661**, 728 (2007); arXiv:0706.3048.

[33] L. Blanchet and C. Skordis, "A new class of relativistic MOND theories," J. Cosmol. Astropart. Phys. **11**, 040 (2024); arXiv:2404.06584.

[34] G. Bianconi, "Gravity from entropy," Phys. Rev. D **111**, 066001 (2025); arXiv:2408.14391.

[35] T. Padmanabhan, "Cosmic information, the cosmological constant and the amplitude of primordial perturbations," Phys. Lett. B **773**, 81 (2017); arXiv:1703.06144.

[36] DESI Collaboration, "DESI 2024 VI: Cosmological constraints from the measurements of baryon acoustic oscillations," arXiv:2404.03002 (2024).

[37] W. Hu, "Lecture notes on CMB theory: From nucleosynthesis to recombination," arXiv:0802.3688 (2008).

[38] M. Haubner, F. Lelli, *et al.*, "BIG-SPARC: A new galaxy sample for testing dark matter and galaxy formation models," arXiv:2411.13329 (2024).

[39] S.-K. Huang, "The arrow of time from Petz recovery" (Paper I), arXiv:2603.xxxxx (2026).

[40] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," Phys. Rev. Lett. **75**, 1260 (1995); gr-qc/9504004.

[41] T. Jacobson, "Entanglement equilibrium and the Einstein equation," Phys. Rev. Lett. **116**, 201101 (2016); arXiv:1505.04753.

[42] L. Smolin, "MOND as a regime of quantum gravity," Phys. Rev. D **96**, 083523 (2017); arXiv:1704.00780.

[43] E. Pazy, "Quantum statistical modified entropic gravity as a theoretical basis for MOND," Phys. Rev. D **87**, 084063 (2013); arXiv:1302.4411.

[44] T. de Paula Netto, L. Modesto, and I. L. Shapiro, "Universal leading quantum correction to the Newton potential," Eur. Phys. J. C **82**, 160 (2022); arXiv:2110.14263.

[45] J. F. Donoghue, "A critique of the asymptotic safety program," Front. Phys. **8**, 56 (2020); arXiv:1911.02967.

[46] E. Deddo, J. Kang, L. Ren, and A. C. Wall, "Entropic irreversibility theorems," J. High Energy Phys. **09**, 179 (2024); arXiv:2404.15077.

[47] D. Grumiller, "Model for gravity at large distances," Phys. Rev. Lett. **105**, 211303 (2010); arXiv:1011.3625.

[48] M. M. Brouwer *et al.*, "First test of Verlinde's theory of emergent gravity using weak gravitational lensing measurements," Mon. Not. R. Astron. Soc. **466**, 2547 (2017); arXiv:1612.03034.

[49] S. De Vuyst, P. A. Höhn, A. Kirklin, and J. Eccles, "Gravitational entropy is observer-dependent," J. High Energy Phys. **07**, 146 (2025).

[50] A. J. Parzygnat and F. Buscemi, "Axioms for retrodiction: Achieving time-reversal symmetry with a prior," Quantum **7**, 1013 (2023).

[51] O. Fawzi and R. Renner, "Quantum conditional mutual information and approximate Markov chains," Commun. Math. Phys. **340**, 575 (2015).

[52] S.-K. Huang, "One equation, three scales: $\Sigma = D(\rho_{\mathrm{spacetime}} \| \rho_{\mathrm{matter}})$ as the universal action," Paper IV in this series (2026).